

Analytic Combinatorics of Unlabeled Objects

Set of exercises 3

October 5, 2026

1. Multivariate Generating Functions

A) Integer compositions*

We recall that integer compositions are specified by

$$\mathcal{C} = \text{Seq}(\{1, 2, 3, \dots\}) \simeq \text{Seq}(\text{Seq}_{\geq 1}(\mathcal{Z})).$$

During class, we used OGFs to prove that the average number of terms in a composition of n is $\sim n/2$.

1. What if we restricted the compositions to have an even number of terms? Find the total number of compositions of n into an even number of terms.
2. Let us consider now the number of even terms in a general composition of n . Show that the average number of even terms is $\sim n/6$.
3. Now we want to fix the number m of even terms. Find the generating function.

B) Binary trees

We consider the binary trees

$$\mathcal{B} = \mathcal{E} + \mathcal{Z} \times \mathcal{B} \times \mathcal{B},$$

where the node $\mathcal{E}$ is the empty-node, while $\mathcal{Z}$ is an atom of weight 1.

1. Give a specification for the binary trees $\mathcal{B}^\circ$ marking the length of the left-most branch.
2. Solve the equation for $B(z, u)$ and show $\partial_u B(z, 1) = z(B(z))^3$.
3. Let $A(z) = zB(z)$. Apply the Lagrange Inversion Formula to compute $[z^n](A(z))^3$.
4. Deduce that the average length of the left-most branch tends to 3 as $n \rightarrow \infty$.
5. Generalize. Find the exact number of binary trees of size n with left-most branch of length m .

2. The Kraft-McMillan inequality

In this exercise we prove a key inequality in Coding and Information Theory by using the radius of convergence.

Let $\mathcal{A} = \{a_1, \dots, a_k\}$ and $\mathcal{B} = \{0, 1\}$. A **binary code** $c: \mathcal{A}^* \rightarrow \mathcal{B}^*$ is a morphism, i.e., it satisfies $c(xy) = c(x)c(y)$. The code c is said to be *uniquely decodable* iff it is injective.

1. Let¹ $A(z) = \sum_{a \in \mathcal{A}} z^{|c(a)|}$. Show $C(z) = \sum_{w \in \mathcal{A}^*} z^{|c(w)|} = \frac{1}{1-A(z)}$.

2. [Kraft-McMillan inequality] Prove that if c is uniquely decodable, then

$$\sum_{a \in \mathcal{A}} 2^{-|c(a)|} \leq 1.$$

Hint. Reason by contradiction. Note that power-series are continuous within their radius of convergence.

3. [Shannon bound] (Optional) Consider a distribution $\mathbf{p} = (p_1, \dots, p_k)$ over $(a_1, \dots, a_k)$. Prove that, for a random symbol X ,

$$H(\mathbf{p}) = \sum_{j=1}^k p_j \log_2(1/p_j) \leq \sum_{j=1}^k p_j |c(a_j)| = \mathbb{E}[c(X)].$$

Hint. Prove that $\log x \leq x - 1$ for all $x \in \mathbb{R}_{>0}$. Use the Kraft-McMillan inequality from part 2.

3. Meromorphic asymptotics*

For each of the following functions $f(z)$, find the *radius of convergence* and an *asymptotic* (equivalent) for the coefficients (a_n) of the power-series expansion $f(z) = \sum a_n z^n$ of $f(z)$ at $z = 0$.

1. $f(z) = \frac{1}{(1-2z)^2} \log\left(\frac{1}{1-z}\right)$.

2. $f(z) = \frac{e^{2z}-1}{(1-z)(1-2z)^3}$.

3. $f(z) = \frac{1}{2-e^z}$.

¹Here $|w|$ is the length of the word, i.e., the number of symbols.